

Introduction to Networks

Q1. Diagram: How a phone connects to Instagram

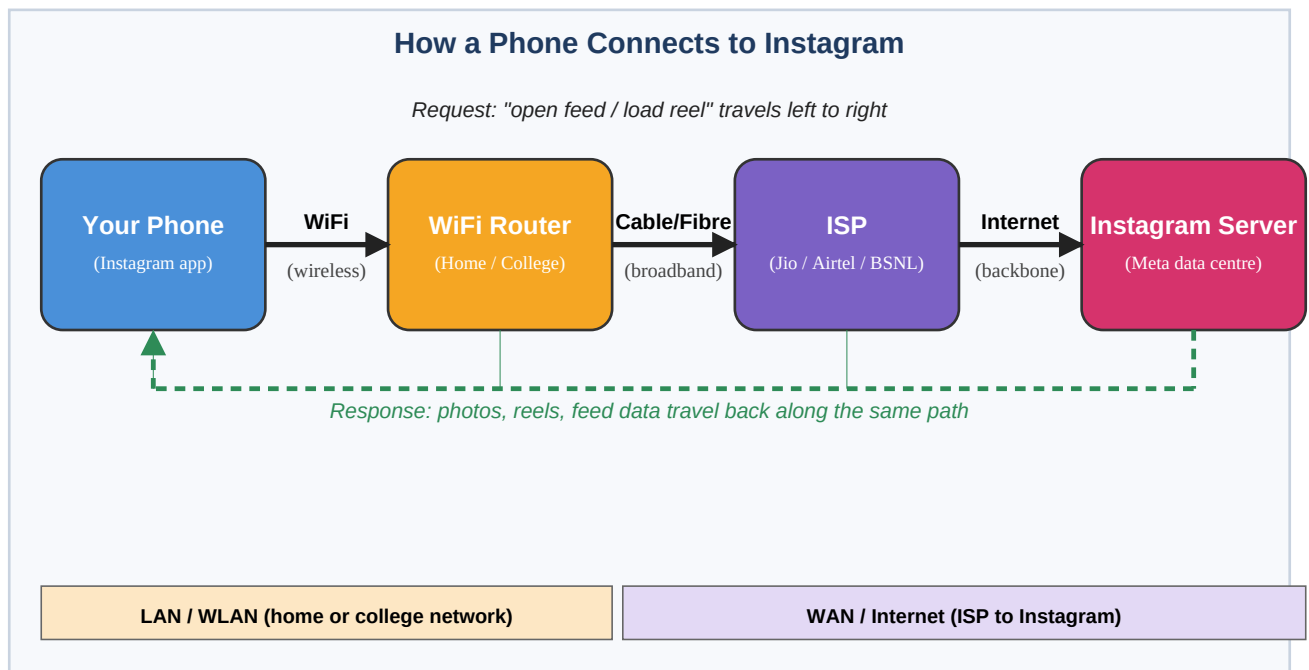


Figure 1: Connection path from phone to Instagram server (request shown left to right, response shown returning)

Part	Label / Role
Phone	Runs the Instagram app and sends requests (view feed, load reel, post photo).
WiFi router	Connects the phone wirelessly to the local network and forwards traffic to the ISP.
ISP	Internet Service Provider that gives access to the wider Internet and routes data onward.
Instagram server	Meta's data centre that stores content, processes the request and sends the response.
Connections	Phone to router: WiFi (wireless). Router to ISP: broadband cable/fibre. ISP to Instagram: Internet backbone links.

Q2. Five everyday devices that are part of a network

No.	Device	Network type	Explanation
1	Smartphone	WLAN (LAN) / WAN	Joins home or college WiFi (LAN); on mobile data it connects to the operator's WAN.
2	Smart TV	LAN (WLAN)	Connects to the home WiFi to stream Netflix or YouTube through the router.
3	WiFi router	LAN and WAN	Creates the local network on one side and links it to the ISP's WAN on the other.
4	Laptop	LAN / WLAN	Connects by WiFi or Ethernet cable to the college or home network.
5	Smartwatch	PAN	Pairs with the phone over Bluetooth within a range of a few metres.

LAN = Local Area Network, WAN = Wide Area Network, PAN = Personal Area Network, WLAN = Wireless LAN.

Q3. Difference between the Internet and an intranet

The **Internet** is a global public network of interconnected networks that anyone with access can use. An **intranet** is a private network limited to one organisation, and only authorised users can reach it, usually from inside the campus or through a login.

Point	Internet	Intranet
Access	Open to everyone worldwide	Only students and staff of the college
Ownership	No single owner; many ISPs and organisations	Owned and managed by one organisation
Content	Websites, social media, videos, shopping	Attendance, timetable, ERP, internal notices
Security	Public, so more exposed to threats	Restricted, protected by firewall and login
Example	Public WiFi at a railway station or cafe used for Instagram or Google	College portal or library system reachable on campus WiFi

Example: On the college WiFi, a student can open the college ERP portal to check attendance and results. This portal lives on the college's intranet and is meant only for its students and staff. The same WiFi also gives Internet access, so the student can open YouTube or Google. On public WiFi at a railway station, the student can browse the Internet but cannot open the college's internal systems, because those are not open to the public.

Q4. How data travels when I search in an app (Zomato)

When I type "pizza" in Zomato and hit **Search**, the app on my **smartphone** creates a request and sends it through the **WiFi router** (LAN) or mobile data to my **ISP**. The ISP routes the request over the **Internet (WAN)** to Zomato's **server**, and a DNS lookup first finds the server's IP address. The server searches its database for nearby restaurants serving pizza and prepares the results (names, ratings, prices, images). This response travels back along the same path: server, ISP, router, phone. The app then displays the restaurant list on the screen, usually within a second.